

Predicting Compressive Strength and Optimal Replacement Level for Rice-Husk-Ash Blended Concrete from a Sparse Literature Dataset

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Abstract

We build a machine-learning pipeline to predict the compressive strength and the optimal cement-replacement level of rice-husk-ash (RHA) blended concrete, using a dataset compiled from **79 research papers**. The central obstacle is data sparsity: strength values are packed inside free-text cells as *(replacement level $\times$ curing age)* grids. Parsing and exploding these cells—rather than any modeling choice—is the highest-leverage step, turning ~ 70 packed rows into **756 clean data points across 44 papers**. Through a five-axis ablation study we find that (i) absolute strength is essentially unpredictable across papers ($R^2 < 0$) because the corpus mixes mortars, normal concrete and ultra-high-performance concrete, but becomes learnable once the paper’s own control (0%) strength is supplied as an input; (ii) a strongly-regularized, monotonically-constrained gradient-boosted model on just three inputs (*replacement, age, control strength*) is the best predictor, reaching $R^2 = 0.50 \pm 0.20$, **MAE ≈ 10 MPa**; and (iii) text embeddings (MiniLM/BERT-style) provide *no reliable gain* at this sample size. The optimal-replacement target cannot yet beat a naive prior—the derived predictor in fact degenerates to a constant. We conclude that the binding constraint is data quantity and the absence of base-mix descriptors, not model capacity.

1 Introduction

Rice husk ash is a pozzolanic by-product that can partially replace Portland cement, reducing both cost and CO₂ footprint. The two engineering quantities of interest are the **compressive strength** of the resulting concrete and the **optimal replacement level**—the percentage of cement that can be replaced by RHA while maximizing strength. Our goal is a predictive model for both, trained on a dataset hand/LLM-extracted from the literature. The practical difficulties are (a) pervasive missing values and (b) selecting a model that performs well on a small, heterogeneous dataset. We take XGBoost as the baseline and interrogate the data and model choices rigorously.

2 The Dataset: What It Actually Contains

The source file `RHA Master sheet v2.csv` contains 225 columns organized under a **four-row hierarchical header**: category ($\rightarrow$ PAPER METADATA/INPUT/OUTPUT), section, feature (55 features), and a sub-field. Every feature is split into four columns—**Extracted Value, Unit, Source Type, Exact Evidence**—of which only the first two carry predictive information; the latter two are provenance and are discarded. The file is `latin-1` encoded.

The dataset contains **79 papers** with unique titles. Sparsity is uneven rather than uniform: among the 44 papers that yield strength data, availability of the 23 retained features ranges from 96% (processing method) and 89% (SiO_2) down to 18% (BET surface), i.e. **5–82% missing**, with 8 of the 23 features more than half missing (Figure 1b). The prediction targets are present in 70 (strength), 71 (optimum), and 76 (replacement levels) papers.

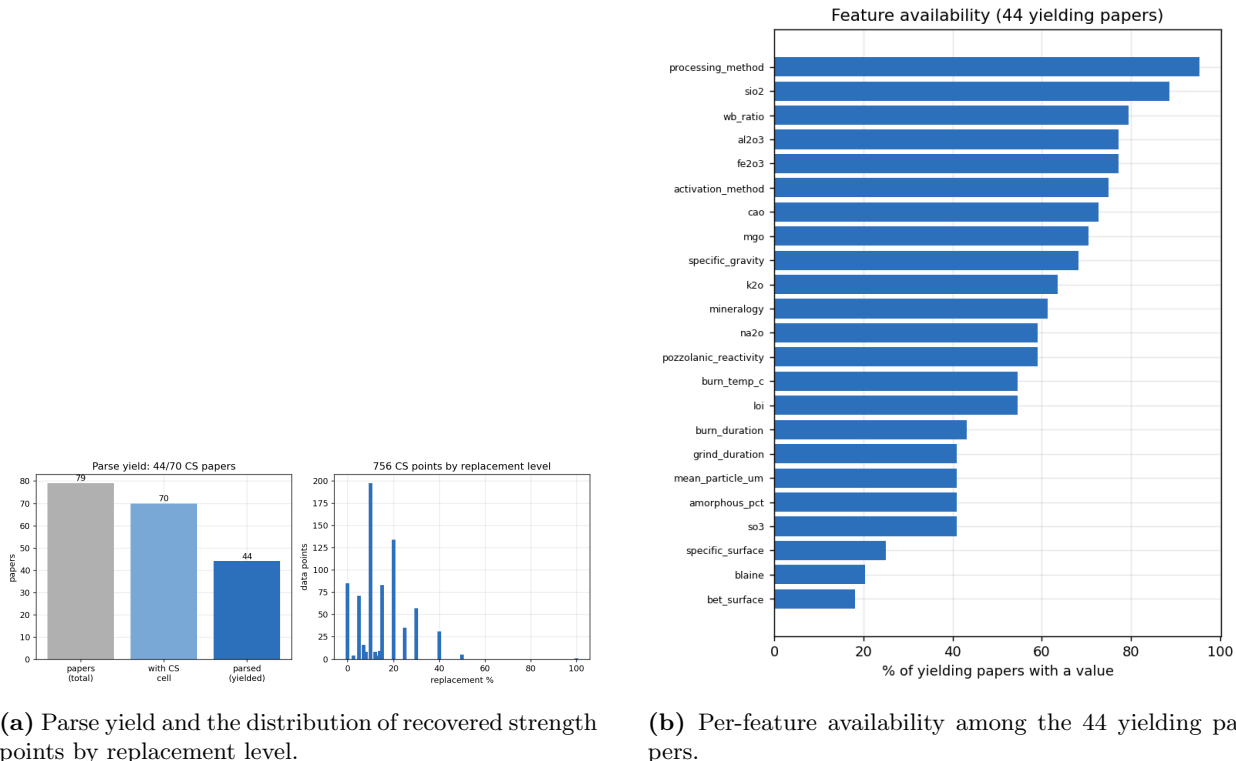


Figure 1: The dataset is small and sparse. Chemistry (SiO_2 , oxides) and w/b are reasonably populated; fineness measures (BET, Blaine, specific surface) are scarce.

3 Data Representation: Parsing the Packed Cells

The single most valuable observation is that each Compressive Strength cell encodes a *grid* of measurements in one of ~ 12 textual layouts, e.g.

3 days: 20 (7%), 20 (10%); 28 days: 44.5 (7%), 46 (10%); ...

We built a state-machine parser (`02_parse_explode.py`) that resolves, for each value, a *curing age* and a *replacement level* from inline qualifiers, segment headers, or persistent line context, and emits a tidy long table with one row per (*paper*, *replacement %*, *age*). Precision safeguards include:

- unsigned number matching, fixing mis-reads such as $\text{RHA}8-10\% \rightarrow 10$ (not -10);
- a **per-paper whitelist**: a resolved replacement % must match the paper’s declared levels (plus 0 for control), which rejects mis-labelled sample IDs ($\text{RHA}1$, $\text{M}2$) and recovers trailing-code mixes (Aa-10 , A35-10);
- physical sanity clamps ($0 \leq \% \leq 100$, $0 < \text{age} \leq 400 \text{ d}$, $0.5 \leq \text{MPa} \leq 250$).

This recovers **756 data points across 44 of the 70 strength papers** (226 at the standard 28-day age). The 26 non-yielding papers are graph-derived narratives, pure ranges, or bare number lists with no age/level mapping—genuinely unrecoverable without the source PDFs, and logged to `parse_failures.csv`.

Figure 2 checks the parsed data against expected physics, but both checks must be made *within* a paper: a raw cross-paper average is dominated by mix type (the corpus spans 1–205 MPa), not by the variable of interest. Normalising each *(paper, replacement level)* series by its own 28-day value, strength does rise with curing age—91% of pre-28-day points fall below the series’ own 28-day value and 82% of later points exceed it, with the median ratio climbing from 0.53 at 1 d to 1.00 at 28 d and 1.21 at 90 d. The strength-vs-replacement shape, by contrast, is *not* reliably unimodal: of the 33 papers reporting at least three distinct replacement levels at 28 d, only 14 (42%) peak at an interior level, while 12 (36%) are highest at the lowest level tested—often the 0% control, meaning RHA never helped in that paper—and 7 (21%) are still rising at the highest level tested. An interior optimum is therefore common but far from universal, which is exactly why the optimum-prediction task in Section 8 is hard.

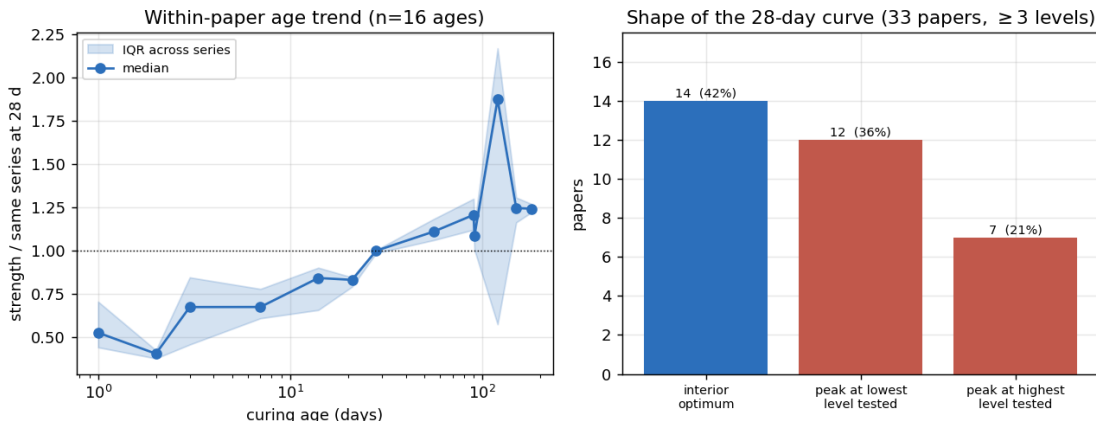


Figure 2: Physics checks on the parsed data, computed within papers. **Left:** each *(paper, replacement level)* series is normalised by its own 28-day strength; the median rises with age (shaded band is the IQR across series). Wiggles at 2, 91 and 120 d come from ages with very few series and are small-*n* noise, not a reversal of the trend. **Right:** the 28-day strength-vs-replacement curve has an interior optimum in only 42% of papers; in 36% the best level tested is the lowest one.

4 Feature Engineering

Numeric features (oxides, fineness, w/b, burn temperature/duration) are parsed as the mean of the numbers in each cell, robustly handling multi-sample entries such as “94.6 (RHA-A), 93.7 (RHA-B)”. High-cardinality but semantically redundant text fields are **canonicalized** into compact categoricals: activation method $\in \{\text{none, thermal, mechanical, both}\}$; mineralogy $\in \{\text{amorphous, mixed, crystalline}\}$; processing $\in \{\text{controlled, uncontrolled, ...}\}$; pozzolanic reactivity as an ordinal. We deliberately apply **no imputation** for the tree models, which handle missing values natively by learning a default split direction; imputers (median/KNN/MICE) are evaluated separately and only ever fit *inside* each cross-validation split to prevent leakage.

5 Methodology

Because rows from the same paper are strongly correlated, all evaluation uses **grouped cross-validation by paper**: no paper appears in both training and test. Given the small sample, a single split is unreliable, so we report the mean and standard deviation of **repeated grouped hold-out** (GroupShuffleSplit, 30% of papers held out, 40 repeats). We compare three target framings:

- A. Absolute strength, all papers** predict MPa directly from RHA features.
- B. Relative-to-control** predict strength normalized by the paper’s 0% control.
- C. Absolute strength given control** predict the RHA-blend strength using the paper’s control (0%) strength as an additional input—always known in practice, hence not leakage. Control rows themselves are excluded from evaluation to avoid a trivial identity.

6 Ablation Study

We ablate five axes. All numbers are repeated grouped hold-out (mean $\pm$ std) over 40 repeats, except the HistGB row of Study 2 and the three imputer rows of Study 4, where 31 repeats survived (a split is discarded when a held-out fold ends up with no usable rows). Comparisons *within* a study are like-for-like unless noted.

Study 1 — Target framing. Absolute strength across papers is *unpredictable* ($R^2 = -0.78$); the corpus mixes mortars, normal concrete and UHPC (1–205 MPa) whose baseline levels are set by mix factors absent from the data. Supplying the control strength (framing C) makes the RHA effect learnable (Table 1).

Table 1: Study 1: target framing (XGBoost, all structured features).

Framing	R^2	std	RMSE	MAE
C: blend control strength	0.395	0.259	16.45	10.64
B: relative-to-control	−0.159	0.300	0.29	0.22
A: absolute, all papers	−0.776	1.549	30.99	20.95

Study 2 — Model. Random Forest and XGBoost dominate; HistGradientBoosting is unstable and linear Ridge fails badly on this tiny, collinear, missing-laden matrix (Table 2). Random Forest posts the higher mean here (0.436 vs 0.395), but the gap is roughly one-sixth of the run-to-run spread of either model ($\sigma \approx 0.21$ –0.26) and so is not a reliable ordering. We nonetheless carry **XGBoost** forward, because the largest single gain in the whole study comes from the monotone constraints of Study 5 ($0.442 \rightarrow 0.482$), which XGBoost supports natively and **RandomForestRegressor** does not. The tuned monotone XGBoost of Section 7 reaches 0.50 ± 0.20 , above the unconstrained Random Forest.

Study 3 — Feature set. Parsimony wins decisively: the three-feature *base* {replacement, age, control} is the best set, and *every* added feature group degrades generalization—clear evidence of overfitting at $n=23$ papers (Table 3, Figure 3).

Table 3: Study 3: feature set (framing C, XGBoost).**Table 2:** Study 2: model (framing C, all features).

Model	R^2	RMSE
Random Forest	0.436	15.82
XGBoost	0.395	16.45
Mean baseline	-0.310	23.97
HistGB	-0.524	22.28
Ridge	-14.7	44.74

Feature set	R^2	RMSE
base [repl, age, control]	0.442	15.54
base + fineness	0.435	15.73
base + categorical	0.427	15.93
all + MiniLM text	0.426	16.13
all + TF-IDF text	0.403	16.42
all structured	0.395	16.45
base + all numeric	0.349	16.85
base + chemistry	0.317	17.18
base + processing	0.236	17.69

Study 4 — Imputation. For the tree models, native NaN handling matches explicit imputation on R^2 : native, median and KNN sit within 0.007 of one another, and only MICE clearly hurts (Table 4). The lower RMSE of the median and KNN rows is *not* evidence that they generalize better—those rows completed 31 of the 40 repeats, so their error is averaged over an easier subset of splits and is not directly comparable to the native row’s 40. Since no imputer improves R^2 and each adds a fitted component that must be re-fit inside every split, we keep **native handling**.

Study 5 — Physics priors. Adding **monotone constraints**—strength non-decreasing in curing age and in control strength—gives the single best result, $R^2 = 0.482 \pm 0.217$, encoding domain knowledge the tiny sample cannot reliably learn on its own (Table 5).

Table 4: Study 4: imputation (framing C).

Scheme	R^2	RMSE
XGB + native	0.397	16.40
XGB + median [†]	0.396	14.88
XGB + KNN [†]	0.390	14.92
XGB + MICE [†]	0.249	16.35

[†]31 of 40 repeats; RMSE not comparable to the native row.

Table 5: Study 5: physics priors (base features).

Prior	R^2	RMSE
monotone(age, control)	0.482	15.02
log + monotone	0.458	15.71
plain	0.442	15.54
log-target	0.423	16.09

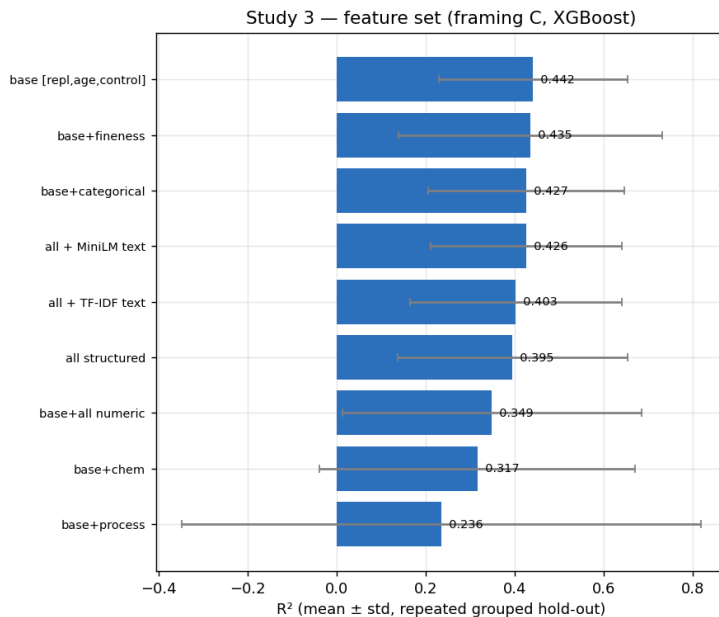


Figure 3: Feature-set ablation: parsimony wins (repeated grouped hold-out, mean $\pm$ std).

7 Final Model and Results

Features used (framing C). The final model takes exactly three inputs—no RHA chemistry, physical, or processing features are included, and Study 3 above is the evidence for why (every added feature group made generalization *worse* at $n = 23$ papers):

Feature	Meaning	Monotone constraint
<code>replacement_pct</code>	RHA cement-replacement level of the blend row (0–100%)	none
<code>age_days</code>	Curing age at which strength is measured (days)	non-decreasing
<code>ctrl_strength</code>	The <i>same paper’s own</i> 0% (control) compressive strength at the matching curing age (MPa)	non-decreasing

The target is `strength_mpa`, the RHA-blend row’s own compressive strength. `ctrl_strength` is looked up per (`paper_id`, `age_days`) from that paper’s own 0%-replacement rows (Section 7 onward always assumes it is known at inference time, since it is the paper’s own OPC baseline, not a property of the RHA blend being predicted). Combining these three features with monotone constraints and a light grouped hyper-parameter search yields a deliberately shallow model (`max_depth=2`, `learning_rate=0.03`, 300 trees). Its performance is

	R^2	RMSE (MPa)	MAE (MPa)
Final model (repeated grouped hold-out)	0.50 ± 0.20	14.9	10.1
Mean baseline	−0.31	24.0	17.4

Figure 4a shows out-of-fold predictions. Note that its panel title reports $R^2 = 0.63$, MAE = 9.4 MPa, which is *not* the headline number above: the scatter comes from a single 5-fold **GroupKFold** pass whose out-of-fold predictions are pooled into one R^2 over all 335 rows, whereas the table reports the mean of 40 grouped hold-outs that each train on fewer papers (70% vs 80%) and are

scored split-by-split. The pooled single-pass figure is the more optimistic of the two; we quote the repeated hold-out as the honest estimate and show the pooled predictions only because they give one prediction per row to plot.

The model tracks normal-strength concrete well but systematically *under-predicts* the high end. Note that the 1–205 MPa span quoted in Study 1 describes framing A; framing C is evaluated on the 335 blend rows of 23 papers that also report a 0% control, and that subset reaches 186 MPa. Within it there is **exactly one** paper above 120 MPa (8 points, 120–186 MPa), and the model’s predictions saturate near 90 MPa there, under-shooting by 70 MPa on average. The corpus does contain a second, even stronger paper—it holds the 205 MPa maximum—but it reports no control rows at all and is therefore excluded from framing C entirely, which is part of why 44 papers reduce to 23 here. With a single high-strength group in the evaluated set, grouped validation always places it wholly in the test fold, so the model has never seen a UHPC example when it is scored—an honest and interpretable failure mode, but one driven by a single group rather than by a subpopulation. That tail is also what pulls the residual distribution (Figure 4b) slightly negative: residuals are centred near zero (mean -2.7 MPa, median -2.2 MPa) but carry a long left tail reaching -100 MPa, all of it from that one paper. Feature importance (Figure 5) confirms that control strength (0.74) and curing age (0.17) carry most of the signal, with replacement level (0.09) modulating on top.

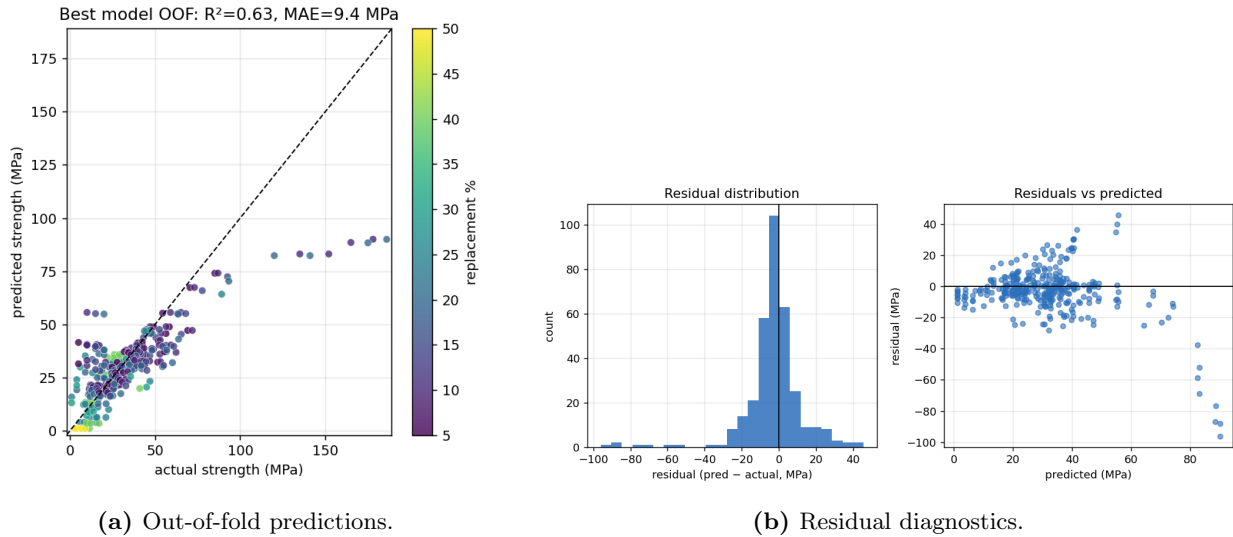


Figure 4: Final model behaviour, from a single pooled 5-fold `GroupKFold` pass ($R^2 = 0.63$ pooled, versus the 0.50 ± 0.20 repeated-hold-out estimate quoted in the table above). Colour encodes replacement level. The dashed line is the ideal $y = x$; the points falling far below it at the right of (a) are the single UHPC paper.

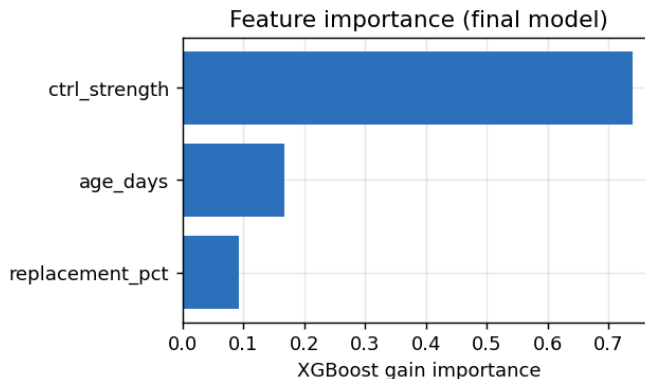


Figure 5: Feature importance of the final model. The paper’s control strength and the curing age dominate; replacement level provides the RHA-specific modulation.

8 Predicting the Optimal Replacement Level

We derive the optimum as the replacement level that maximizes the predicted 28-day strength curve of the final model, evaluated on held-out papers, and also test a direct per-paper regressor. Neither beats simply predicting the corpus-mean optimum: on the 20 held-out papers with a reported optimum the argmax approach gives MAE = 4.75 percentage points against a mean-baseline of 3.45, and the direct regressor gives 6.5 against its own baseline of 5.3 over 43 papers.

Figure 6 shows why, and the failure is starker than “fails to beat the baseline” suggests: **the argmax model returns the same value, 7.5%, for every one of the 21 held-out papers.** It is a constant predictor, and a miscalibrated one, since the corpus mean is 11.5%. This is a direct consequence of the final model’s structure—it is monotone and non-decreasing in control strength and age, and with `max_depth=2` there is almost no interaction between replacement level and the other two inputs, so changing a paper’s control strength shifts the predicted strength curve vertically without moving the location of its maximum. Every paper therefore inherits the same argmax.

The reported optima it is scored against are themselves concentrated but not tight: 75% of papers fall in 10–15% (mean 11.5%, median 10%), yet the full range runs 0–20% with a standard deviation of 4.6 points. A constant-11.5% prior already achieves MAE 3.45, so beating it requires resolving paper-to-paper shifts of a few percentage points—and with 20–43 labelled papers, and an interior optimum present in only 42% of them (Section 3), that signal is not available. Predicting the optimum is best treated as an open problem on this corpus rather than a solved output.

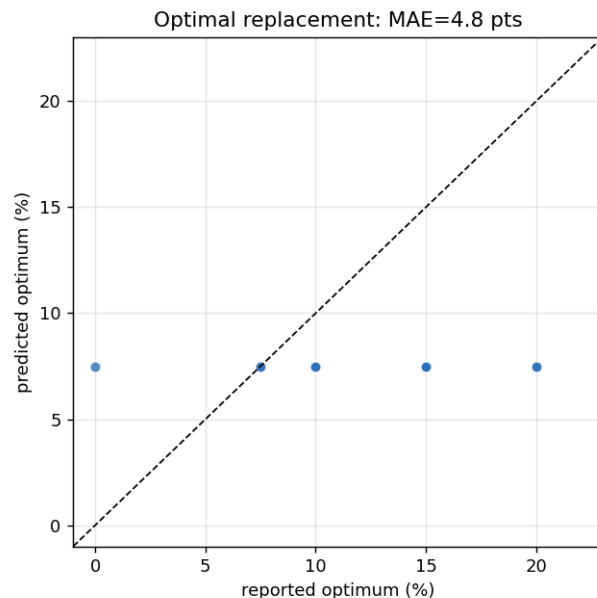


Figure 6: Predicted vs reported optimal replacement level. The predictions form a horizontal line: the model emits a constant 7.5% for every held-out paper, below the 11.5% corpus mean, and so loses to the mean prior it is compared against.

Part II — Sensitivity to the UHPC-Regime Outliers

Sections 7–8 flagged, but did not remove, the two papers responsible for the corpus’s 1–205 MPa span. This part removes them and re-measures both target framings, to separate “the model degrades on UHPC” from “the model is fine once UHPC is taken out of scope.”

9 Which Papers, and Why

The cutoff is applied at the *paper* level, not the row level: a paper is dropped in full—every replacement level, every curing age—if its **maximum** reported compressive strength exceeds **120 MPa**. This threshold is not a round-number guess; it sits in a genuine gap in the data. Of the 44 papers, only two ever cross it, and the next-highest paper after them tops out at exactly 100.0 MPa (Table 6).

ID	Title (abridged)	Max str.	Why it is out of scope
17	“...to produce ultra high performance concrete”	205.5 MPa	Explicitly UHPC. $w/b = 0.15\text{--}0.23$ vs. $\approx 0.35\text{--}0.50$ for the rest of the corpus. Strength rises smoothly 51 $\rightarrow$ 205.5 MPa with age—internally consistent, just a different mix-design regime (silica fume / superplasticizer / very low w/b), not ordinary RHA concrete.
36	“Analysis of the effects of RHA on hydration...”	186.0 MPa	Reports 125–135 MPa at 1 day of curing, including for its own 0% control—not physically plausible at any w/b . Every value is flagged <code>is_range=True</code> (read off a graph axis, not a table), consistent with a mis-read scale rather than genuine UHPC.

Table 6: The two papers removed by the 120 MPa cutoff, and the highest strength reported by any of the 42 papers kept: 100.0 MPa.

This directly tests the diagnosis already made in Section 7: that framing A’s negative R^2 comes from mixing mortars, normal concrete and UHPC on one strength scale, and that framing C is largely insulated from this because it conditions on each paper’s own control strength. Paper 17 was never in framing C to begin with (it reports no 0% control); paper 36 was one of its 23 papers.

Reproducibility check. Both sides below were re-run with the identical model class and hyperparameter grid as Sections 7–8. The unfiltered “before” run here reproduces the Section 7 headline almost exactly ($R^2 = 0.503 \pm 0.197$, RMSE= 14.90, MAE= 10.02 vs. the original $R^2 = 0.50 \pm 0.20$, RMSE= 14.9, MAE= 10.1), confirming the numbers below are directly comparable to the rest of this report.

10 Results With and Without the UHPC Papers

Framing A (absolute strength, no baseline info)	R^2	RMSE (MPa)	MAE (MPa)
With UHPC papers (44 papers, 756 rows, 1–205.5 MPa)	-0.43 ± 0.78	29.5	20.2
Without UHPC papers (42 papers, 706 rows, 1–100 MPa)	$+0.08 \pm 0.23$	17.1	13.0

Framing C (tuned, blend control)	R^2	RMSE (MPa)	MAE (MPa)	OOF R^2
With UHPC papers (23 papers, 335 rows)	0.50 ± 0.20	14.9	10.0	0.63
Without UHPC papers (22 papers, 327 rows)	0.52 ± 0.16	11.4	8.6	0.60

Optimal replacement (argmax, 28-day, held-out papers)	MAE (pts)	Mean-baseline (pts)
With UHPC papers (20 papers)	4.75	3.45
Without UHPC papers (20 papers)	4.75	3.45

11 Discussion

The two framings respond differently, and the difference is diagnostic rather than incidental. **Framing A flips sign:** R^2 crosses from -0.43 to $+0.08$, and RMSE/MAE fall by 42% and 36% respec-

tively (29.5→17.1 MPa; 20.2→13.0 MPa). A model that is never told a paper is UHPC has no way to know it should predict 180 MPa instead of 40 MPa for a nominally similar input row; once the two UHPC-regime papers are removed, the remaining 42 papers occupy one coherent strength scale (1–100 MPa) and the same features that were noise before now carry signal. $R^2 = 0.08$ is still weak in absolute terms—framing A was never going to become the recommended model—but the sign flip alone confirms the diagnosis.

Framing C improves modestly but genuinely: R^2 rises from 0.50 to 0.52 with a visibly tighter spread across repeats ($\pm 0.20 \rightarrow \pm 0.16$), and RMSE/MAE both drop by double digits (14.9→11.4 MPa, −24%; 10.0→8.6 MPa, −14%). This is smaller than framing A’s swing, exactly as Section 7 predicted: conditioning on the paper’s own control strength already absorbs most of the cross-paper scale variation that framing A is exposed to raw, so removing the outlier papers here mostly trims a noisy, hard-to-fit paper (36) from a 23-paper set rather than unlocking new signal. One number moves the other way—the pooled single-pass OOF R^2 used for the prediction-scatter figure drops slightly ($0.63 \rightarrow 0.60$)—which is expected: that figure is a single 5-fold split, more sensitive to exactly which papers land in which fold when there are only 22–23 of them, and is not the metric either number in the table above is drawn from. The optimal-replacement argmax MAE is *exactly* unchanged (4.75 points against an unchanged 3.45-point baseline, both sides): neither dropped paper had a usable reported optimum, so that evaluation set of 20 papers is identical before and after.

The practical conclusion is that **outlier removal and the control-strength reframing are two different fixes for the same underlying problem**, and only one of them is needed if framing C is what will be deployed. Framing A’s improvement after cleaning is useful mainly as confirmation that the corpus is well-behaved once UHPC is excluded, not as a replacement for framing C, which remains the stronger predictor either way (0.50–0.52 R^2 vs. 0.08 at best). If the eventual goal includes UHPC-range predictions, the two removed papers should be re-added as a separate, explicitly labelled regime (e.g. a categorical “mix class” feature) rather than silently dropped, since 8–38 data points is too little to train a UHPC-specific model on its own; for the conventional-concrete regime this report otherwise targets, dropping them is the right call.

12 An LLM Ablation: Does Control Strength Matter Across Model Families?

Sections 7 and Study 1 (Table 1) established, with a trained regressor, that supplying a paper’s own 0% control strength is what makes RHA-blend strength learnable at all. Two independent LLM experiments ask the same question with no gradient-boosted model in the loop at all: given the same tabular rows, can a general-purpose LLM infer compressive strength from the paper’s inputs alone, and does it too depend on being handed the control strength?

Setup. Six papers were held out at the *paper* level—P036, P022, P007, P030, P054 and P064, none seen in training—giving a fixed benchmark of **57 test points** shared identically by both conditions and both models, so the ablation isolates one variable:

WITH_CONTROL the model is given the paper’s 28-day 0% control strength as an input feature, exactly as framing C does for the ML model.

NO_CONTROL control strength is withheld; only the RHA/mix inputs remain.

Two models. **Claude Fable** was evaluated externally (results as reported); **gpt-oss:120b-cloud** (OpenAI’s open-weight ~ 117 B-parameter model, MXFP4-quantized, accessed through Ollama’s cloud proxy) was run for this report directly against the identical ground truth, one Ollama call per test paper per condition (12 calls total), each given that paper’s held-out rows plus the *entire* condition’s training CSV as in-context reference data, prompted to return a strict JSON array of predictions. gpt-oss was reached only after two earlier choices failed: DeepSeek-R1 8B run locally took 10+ minutes for a single call at this context length on an 8 GB laptop GPU (impractical for 12 calls), and **kimi-k2.6:cloud** returned 402 Payment Required (a paid ollama.com plan). gpt-oss:120b-cloud is free-tier accessible, cloud-hosted, and answered every call in 20–100 s. One call (NO_CONTROL, paper P007) initially degenerated—the model echoed the four *age* values back verbatim instead of predicting strength; it was retried once and produced a normal, non-degenerate answer, which is what is reported below. Because the two models were evaluated under different prompting protocols (Fable’s exact prompt is not available to us), this is not a controlled head-to-head between Fable and gpt-oss—it is two *separate* within-model ablations that happen to share the same benchmark, and both are compared only against themselves (WITH_ vs NO_CONTROL), not against each other.

12.1 Overall Result

Condition	R^2	RMSE (MPa)	MAE (MPa)	Bias (MPa)
Fable, WITH_CONTROL	0.877	8.79	5.84	−2.21
Fable, NO_CONTROL	0.392	19.50	12.83	−6.25
gpt-oss:120b, WITH_CONTROL	0.887	8.41	7.17	−1.48
gpt-oss:120b, NO_CONTROL	0.487	17.92	12.78	−6.82
Final ML model (Section 7, for reference)	0.50 ± 0.20	14.9	10.1	—

Both LLMs move the same way. For Fable, control strength lifts R^2 by +0.485 and cuts RMSE/MAE by roughly **55%**. For gpt-oss, control strength lifts R^2 by +0.400 and cuts RMSE by **53%** (17.92→8.41 MPa) and MAE by **44%** (12.78→7.17 MPa)—the same qualitative effect Study 1 found for the ML model, now reproduced independently by *two* completely different LLMs. The ML row is included only as context, *not* as a matched comparison: it is a repeated grouped hold-out averaged over 40 different 70/30 paper splits, whereas all four LLM rows are single evaluations on one fixed six-paper split. Neither LLM’s WITH_CONTROL number should be read as “beats the ML model”—the honest claim in both cases is the ablation *within* that model, which needs no such caveat because both conditions share the identical test papers.

12.2 Result by Compressive-Strength Category

Each test paper is assigned to one of three strength categories by its own 28-day control strength (Low < 30 MPa: P036, P022; Middle 30–60 MPa: P007, P030; High $\geq$ 60 MPa: P054, P064). The category is a property of the *paper*, not of each individual point—P054’s points, for example, run from 42 to 78 MPa—and it splits the 57 test points 24/18/15. It reveals where control strength actually earns its keep, for both models.

Regime (N)	NO_CONTROL (MAE / Bias, MPa)				WITH_CONTROL (MAE / Bias, MPa)			
	Fable		gpt-oss		Fable		gpt-oss	
	MAE	Bias	MAE	Bias	MAE	Bias	MAE	Bias
Low, < 30 (24)	7.31	+7.31	7.02	+5.86	3.84	+3.70	5.59	+5.26
Mid, 30–60 (18)	3.19	−1.87	5.21	−3.51	3.85	−2.89	7.51	−4.14
High, ≥ 60 (15)	33.22	−33.22	31.09	−31.09	11.43	−10.85	9.30	−9.09

The pattern is sharp, matches the UHPC discussion of Section 9, and now holds for *both* LLMs independently: **without** control strength, both models are reasonably behaved in the low and middle regimes but collapse on the high regime, under-predicting by a mean of 31–33 MPa—neither model has any way to know a given paper’s own strength ceiling, so both regress toward the corpus-typical range. **With** control strength, high-regime MAE falls to 9.30–11.43 MPa (a ~ 65 –70% reduction) because the model is now told, explicitly, which papers are high-baseline. The low and middle regimes—already easy without any baseline signal—show smaller, noisier shifts (gpt-oss’s mid-regime MAE actually rises slightly, $5.21 \rightarrow 7.51$, well within the noise of a 15–18-point evaluation). This mirrors exactly why framing C outperforms framing A for the XGBoost model: the hard part of this corpus was never the RHA effect itself, it is locating each paper’s baseline strength level, and *any* sufficiently capable predictor—tree-based, or either of two unrelated LLM families—needs that baseline supplied to do well outside the middle of the strength range.

12.3 Where It Still Fails: Paper P064

Both models and both conditions agree on the single hardest case. Paper P064 (6 points, high replacement/high strength) is the worst-or-tied-worst per-paper error everywhere:

Paper	MAE, NO_CONTROL		MAE, WITH_CONTROL	
	Fable	gpt-oss	Fable	gpt-oss
P064 (hardest)	47.95	33.70	21.37	12.89
next-worst	23.40 (P054)	29.34 (P054)	4.81 (P054)	7.68 (P007)

Control strength shrinks P064’s error substantially for both models (Fable $47.95 \rightarrow 21.37$; gpt-oss $33.70 \rightarrow 12.89$), but it remains the largest per-paper error by a wide margin in every configuration. At 28 days, NO_CONTROL predictions undershoot this paper’s true 90–111 MPa range by 60–70 MPa (Fable) or land in the 40s-to-60s (gpt-oss)—even with age and replacement given, neither model can guess this paper’s unusually high strength scale without being told it. This is the same failure mode already documented for the final ML model in Section 7 (systematic under-prediction at the high end, worst on the corpus’s most extreme paper)—further evidence that the limitation is in the *data* (a handful of unusually strong papers with no comparable neighbors to generalize from), not specific to any one modeling approach.

12.4 Summary

Two unrelated LLMs, evaluated independently and under different prompting protocols, replicate this report’s central empirical claim, first established with a three-feature, monotone, depth-2 XGBoost tree: **a paper’s own control strength is the dominant signal for predicting RHA-blend compressive strength, and its absence is felt almost entirely at the high-strength end of the corpus.** That three completely different predictors—a gradient-boosted tree and two general-purpose LLMs from different vendors—land on the same qualitative conclusion,

on the same held-out papers, is considerably stronger evidence for that claim than any single result alone.

Part III — Reproducibility of the LLM Pipeline

Part II reported each LLM result from a single run. This part asks whether those numbers can be trusted, through two questions on the same six-paper benchmark:

1. **Scope.** If every data point above 60 MPa is removed—the high-strength regime we are not interested in—do the results improve?
2. **Reproducibility.** If the *same* prompt is sent repeatedly at temperature 0, does the model return the same predictions, and how much do R^2 , RMSE and MAE move from run to run?

Two models were planned for the reproducibility test: **gpt-oss:120b-cloud** (cloud-hosted) and **DeepSeek-R1 8B** (run locally). The first completed; the second could not be run on the available hardware, which is documented in Section 15.2.

13 Scope Change: Removing Points Above 60 MPa

The filter is applied *point by point*: every row whose compressive strength exceeds 60 MPa is removed from both the training files and the test/ground-truth files.

File	Before	After	Removed
Test points (both conditions)	57	49	8: P007 (−1), P054 (−3), P064 (−4)
Training rows, WITH_CONTROL	215	193	22
Training rows, NO_CONTROL	566	501	65

All six test papers still contribute points (P064 is down to two). The true strengths now span 3.1–59.0 MPa instead of 3.1–111.3 MPa. Because the strength categories of Part II are assigned per paper by control strength (Section 12.2), the two high-control papers P054 and P064 still contribute 8 points, all of which are themselves ≤ 60 MPa. In the tables below this category is therefore labelled *high-baseline papers* rather than “high strength”.

14 Does Removing High-Strength Points Improve the Results?

To isolate the effect of the filter, we re-scored the *same* gpt-oss predictions from Part II on the 49 remaining points. No model was re-run, so the predictions are identical and the only difference is which points are counted.

Condition	Points	R^2	RMSE (MPa)	MAE (MPa)	Bias (MPa)
WITH_CONTROL	all 57	0.887	8.41	7.17	−1.48
	≤ 60 MPa (49)	0.797	7.49	6.64	−0.09
NO_CONTROL	all 57	0.487	17.92	12.78	−6.82
	≤ 60 MPa (49)	0.295	13.94	9.79	−2.86

The answer depends on the metric. **The errors improve:** with control strength, RMSE falls by 11% and MAE by 7%, and the bias almost vanishes ($-1.48 \rightarrow -0.09$ MPa); without it, RMSE and MAE both fall by 22–23%. R^2 **gets worse** ($0.887 \rightarrow 0.797$ and $0.487 \rightarrow 0.295$).

Both statements are true at once because R^2 is a *relative* measure:

$$R^2 = 1 - \frac{\text{SS}_{\text{res}}}{\text{SS}_{\text{tot}}},$$

where SS_{res} is the sum of squared prediction errors and SS_{tot} is the total spread of the true values around their mean. The eight removed points were the most extreme in the test set, so removing them shrinks the spread far more than it shrinks the errors:

	All 57	≤ 60 MPa (49)	Change
Standard deviation of true strength	25.0 MPa	16.6 MPa	
SS_{tot} (spread to explain)	35,662	13,512	−62%
SS_{res} , WITH_CONTROL	4,032	2,747	−32%
SS_{res} , NO_CONTROL	18,307	9,523	−48%

With less variation left to explain, even smaller errors explain a smaller share of it. R^2 values computed on the 57-point and 49-point sets therefore should not be compared directly; RMSE and MAE, which are in MPa, are the right measures of whether the model got more accurate. By those measures the filtered benchmark is easier, and the model is more accurate on it—but that is a narrowing of scope, not a free improvement: the model has simply stopped being asked about the points it was worst at. Fable cannot be re-scored the same way, because only aggregate metrics per paper-level category are available for it, not its individual predictions.

15 Reproducibility at Temperature 0

Protocol. Part II sent one prompt per test paper. To test whether *literally the same* prompt gives the same answer, Part III uses one fixed prompt per condition, containing the filtered training CSV and all 49 filtered test rows (each row carrying its own paper ID), and asks for a JSON array of 49 predictions. The prompt is 109,021 characters (≈ 31 k tokens) for WITH_CONTROL and 228,336 characters (≈ 65 k tokens) for NO_CONTROL. Each prompt was sent five times, unchanged, with `temperature = 0` and a fixed `seed = 0`. Temperature 0 means the model always picks its single most likely next token instead of sampling, which is the setting normally used when repeatable output is wanted. Predictions were scored against the ground truth matched by paper, replacement level and age.

15.1 gpt-oss:120b-cloud

Every call returned in 30–52 s. Nine of the ten runs returned a valid answer; one WITH_CONTROL run returned 46 numbers for 49 queries, which cannot be matched to the queries, and is excluded.

Run	WITH.CONTROL			NO.CONTROL		
	R^2	RMSE	MAE	R^2	RMSE	MAE
1	<i>invalid output (46 of 49 values)</i>			0.449	12.32	8.06
2	0.836	6.73	5.55	0.365	13.23	8.20
3	0.380	13.08	10.48	0.069	16.02	11.21
4	0.501	11.74	8.96	0.019	16.45	12.80
5	0.584	10.71	6.90	0.443	12.39	8.92
Mean $\pm$ SD	0.575 ± 0.167	10.56 ± 2.37	7.97 ± 1.89	0.269 ± 0.187	14.08 ± 1.79	9.84 ± 1.86
Range of R^2	0.380 – 0.836			0.019 – 0.449		

The output is not reproducible. No two runs produced the same set of predictions (4 distinct answers from 4 valid WITH_CONTROL runs, 5 from 5 NO_CONTROL runs), and *none* of the 49 queries received the same value in every run. A typical query’s prediction moves by about 4–6 MPa between runs (median standard deviation 5.9 MPa with control, 4.4 MPa without), and the worst by 12–16 MPa. As a result, the same prompt can score anywhere from $R^2 = 0.38$ to 0.84 with control strength.

The usual explanation is how cloud inference is served; we cannot inspect the server, so this is the documented mechanism rather than something we measured. Temperature 0 removes randomness in *choosing* a token, but the probabilities it chooses from are themselves computed on shared GPUs that process many users’ requests together in batches. The order in which the GPU adds numbers depends on what else is in the batch, and floating-point rounding makes that order matter slightly. gpt-oss is also a mixture-of-experts model, in which each token is sent to a subset of sub-networks, so a tiny numerical difference can change which sub-network handles it. Once a single token differs, everything generated after it differs too. The fixed seed does not help, because it controls sampling, which temperature 0 already switches off.

Result by strength category. Mean MAE across the valid runs, with its standard deviation across runs, and the mean bias:

Category (points)	NO.CONTROL		WITH.CONTROL	
	MAE	Bias	MAE	Bias
Low, control < 30 (24)	6.02 ± 3.00	+4.64	5.00 ± 3.00	+3.76
Middle, control 30–60 (17)	7.55 ± 2.94	−3.02	10.78 ± 3.75	−6.64
High-baseline papers, control ≥ 60 (8)	26.15 ± 2.88	−26.15	10.91 ± 5.11	−9.53

Two conclusions from Part II survive the run-to-run noise, and one does not generalise:

- **Control strength still helps overall.** Mean R^2 is 0.575 with control against 0.269 without, and three of the four valid WITH_CONTROL runs score above the best NO_CONTROL run (0.449). The gap in overall MAE (7.97 vs 9.84 MPa) is, however, about the size of a single run’s spread (≈ 1.9 MPa), so it should not be quoted from one run.
- **The benefit is concentrated in the high-baseline papers.** Without control strength, their points are under-predicted by 26 MPa on average even though every one of them is now ≤ 60 MPa; with control strength the error drops to 10.9 MPa. This gap is several times larger than the run-to-run spread, so it is robust.
- **For middle-category papers, control strength did not help in this protocol:** MAE is higher with it (10.78 vs 7.55 MPa), with the predictions pulled low (bias −6.64 MPa).

Part II’s per-paper prompts did not show this, so it may come from mixing all six papers into one prompt; with one Part II run and four Part III runs, we cannot tell.

What this means for Part II. Part II’s single-run value on the same 49 points ($R^2 = 0.797$ with control, Section 14) lies inside this run-to-run range, and is best read as one draw from it rather than as a fixed property of the model. Part II used one prompt per paper and Part III one prompt for all papers, so the difference between 0.797 and the Part III mean of 0.575 mixes the change of prompt with run-to-run noise and cannot be split between them with a single Part II run. The same caution applies to the Fable numbers in Part II, which also come from single runs. **Any LLM result on this benchmark should be reported as a mean and standard deviation over repeated runs.**

15.2 DeepSeek-R1 8B: Not Feasible on the Available Hardware

DeepSeek-R1 8B was to be the local counterpart to gpt-oss: a small model with no cloud dependency and, running on one machine with no other users, potentially more repeatable. It could not complete a single prediction call for this benchmark.

Hardware and model		
GPU	NVIDIA GeForce RTX 5060 Laptop GPU, 8 GB VRAM	
Model	deepseek-r1:8b via Ollama 0.33.1 (Qwen3-8B base, 8.2B parameters)	
Quantisation / size on disk	4-bit (Q4_K_M), 5.2 GB	
Native context window	32,768 tokens (extendable to 131,072)	

Test	Context setting	Outcome
Control: short filler prompt (6,817 tokens)	8,192	Finished in 19.5 s (reading 2,279 tokens/s, writing 51.5 tokens/s)
Part II prompt, one paper (9 rows, 38,722 tokens)	32,768	Rejected: prompt longer than the context window
Same prompt	49,152	No answer after >10 min; stopped
Part III prompt, all 49 rows ($\approx$ 31k tokens)	49,152	No answer within the 15 min limit
Part III prompt cut to 2 rows (same training data, $\approx$ 19k tokens)	49,152	No answer within the 10 min limit

The NO_CONTROL prompt, at about twice the length, was not attempted.

Why. Ollama’s process monitor (ollama ps) during these runs showed the model occupying **12 GB at the 49,152-token setting, split 53% CPU / 47% GPU**. Besides its weights, a language model keeps a working memory (the key-value cache) for every token of the prompt, and this grows with the context length. At about 49k tokens, weights plus working memory need 12 GB, which does not fit in 8 GB of VRAM, so Ollama runs roughly half of the model on the CPU, which is far slower. The short control prompt, which fits on the GPU, ran at normal speed. Cutting the queries from 49 to 2 did not help, which shows that the cost comes from the length of the training data in the prompt, not from the number of predictions asked for. DeepSeek-R1 also writes a hidden reasoning section before its answer, which lengthens every response; how long it ran in these calls is unknown, because no call returned.

Cost of the planned run. Five repeats in two conditions would be ten calls. Each of the calls above ran for 10–15 min without an answer, so the full run would take at least 2.5 hours with no assurance that any call finishes, and the NO_CONTROL prompt is twice as long. The run would become feasible on a GPU large enough to hold the 12 GB WITH_CONTROL footprint entirely (and more for NO_CONTROL), or with a training context short enough to fit in 8 GB (roughly 30–40 rows instead of 193 and 501). The second option is a different, weaker experiment whose results could not be compared with gpt-oss, so no DeepSeek-R1 8B results are reported.

16 Part III Summary

- **Removing points above 60 MPa**, which cuts the test set from 57 to 49 points, lowers absolute errors: MAE falls from 7.17 to 6.64 MPa with control and from 12.78 to 9.79 MPa without. It also lowers R^2 (0.887 to 0.797 with control; 0.487 to 0.295 without), because the spread of true values shrinks by 62%. It narrows the question rather than improving the model, and R^2 is not comparable across the two sets.
- **gpt-oss:120b-cloud is not reproducible at temperature 0.** The same prompt gave a different answer on every run; R^2 with control ranged from 0.38 to 0.84 (mean 0.575 ± 0.167), and 0.269 ± 0.187 without. Temperature 0 does not make a cloud-served model deterministic.
- **The control-strength effect survives this noise**, mainly through the high-baseline papers (MAE 26.2→10.9 MPa), but single-run LLM numbers, including Part II’s, should be treated as one draw from a distribution.
- **DeepSeek-R1 8B could not be run:** with the training data in the prompt, the model needs 12 GB, does not fit the 8 GB laptop GPU, and falls back to 53% CPU execution; no call finished within 10–15 minutes.